

Comprehensive Multivitamin Profile – Laboratory Report (DEMO)

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Patient Name	Sample Patient	Patient ID	MTV2026DEM002
Date of Birth	14Aug1992	Sex	Female
Referring Physician	Dr. Sample Physician, MD	Sample Type	Serum
Report Date	20Jan2026	Collection Time	08:20 AM (Fasting)

1. Summary of Results

Analyte	Result	Reference Range	Flag	Interpretation
Vitamin A (Retinol)	79 µg/dL	30–80 µg/dL	Borderline High	Highnormal; consider supplement review if on retinoids
Vitamin B1 (Thiamine)	112 nmol/L	70–180 nmol/L	Normal	Within reference interval
Vitamin B2 (Riboflavin)	17 µg/L	5–50 µg/L	Normal	Within reference interval
Vitamin B3 (Niacin)	4.3 µg/mL	0.5–8.5 µg/mL	Normal	Within reference interval
Vitamin B6 (PLP)	65 ng/mL	5–50 ng/mL	High	Likely supplementrelated; monitor for neuropathic symptoms if persistent
Vitamin B9 (Folate)	2.8 ng/mL	3–17 ng/mL	Low	Suggest dietary assessment; consider supplementation
Vitamin B12	205 pg/mL	200–900 pg/mL	Borderline Low	Correlate with symptoms; consider MMA/homocysteine if indicated
Vitamin C	0.40 mg/dL	0.4–2.0 mg/dL	Borderline Low	Encourage dietary citrus/vegetables; recheck if symptomatic
Vitamin D (25Hydroxy)	29 ng/mL	30–100 ng/mL	Borderline Low	Insufficient; consider sun exposure and cholecalciferol as per clinician
Vitamin E	18.0 mg/L	5–18 mg/L	Borderline High	Upper limit of normal; review fat-soluble vitamin intake
Vitamin K1	0.18 ng/mL	0.10–2.20 ng/mL	Normal	Within reference interval

2. Clinical Interpretation & Commentary

Borderline/Low Findings: Vitamin D is insufficient (29 ng/mL). Vitamin B12 and Vitamin C are borderline low, and folate is low. These values may reflect inadequate dietary intake, low sun exposure, or malabsorption. Confirmatory testing such as methylmalonic acid or homocysteine may help evaluate functional B12 deficiency.

High Values: Vitamin B6 is elevated above the physiological range, typically associated with excessive supplementation. Persistent high levels may cause neuropathic symptoms and require dosage review. Vitamins A and E show upper-range values and should be monitored in patients using fat-soluble vitamin supplements.

3. Methodology

Fat-soluble vitamins (A, D, E, K)	High-Performance Liquid Chromatography (HPLC)
Water-soluble B-complex vitamins	LC-MS/MS (Liquid Chromatography–Tandem Mass Spectrometry)
Vitamin C	Spectrophotometric Analysis

4. Notes

Reference ranges may vary between laboratories depending on assay methodology and population differences. Results should always be interpreted in conjunction with clinical symptoms, medical history, dietary intake, and physician assessment. This is a **dummy demonstration report** and is not intended for diagnostic or therapeutic purposes.